HyperText Markup Language

HTML is a markup language used to create web pages. HyperText refers to text that contains hyperlinks, which connect one web page to another.

Structure of an HTML document

A Document written in HTML should contain the **.html** extension.

```
<!DOCTYPE html>
<html lang="en">
      <head>
            <meta charset="UTF-8">
            <meta name="viewport" content="width=device-width,
            initial-scale=1.0">
            <title>Page Title</title>
      </head>
      <body>
      </body>
</html>
```

Tags

Tags are keywords enclosed in angle brackets (**<>**) used to create HTML elements.

Attributes

Attributes are used to define special properties for an HTML element.

```
<opening_tag attribute_name="value">Content</closing_tag>
```

<!DOCTYPE html>

declares the document as an HTML5 document.

<html>

This is the root element of the webpage. The `lang` attribute specifies the language of the webpage.

<head>

The Head section contains metadata(information) about the webpage.

<meta>

used to specify metadata about the HTML document.

```
<meta name="" content="">
```

<meta charset="UTF-8">

sets the character encoding to UTF-8, which supports a wide range of characters.

Viewport

It is the visible area of the website. Set to the width of the device with no zoom.

<title>

sets the title of the webpage on the head.

<body>

The Body section contains the content of the webpage.

HTML validator - https://validator.w3.org/

Head Section

<link>

The <link> tag is used to link an external file to an HTML document.

Attribute	Description
rel	specifies the relationship between the HTML document and the linked file.
href	specifies the path or location of the linked file.
type	specifies the type of the linked file.

Creating a Favicon

```
<link rel="icon" href="file_path" type="image/file_format">
```

HTML Text

Element	Description
<h1></h1>, <h2></h2>,…<h6></h6>	Heading
	Paragraph
<hr>	Horizontal Rule
 	Line break
(Entity)	Blank space

White Space Collapsing

A behavior in HTML where the browser combines multiple white spaces(space, tabs, and newlines) into a single space while displaying content on a webpage.

HTML Lists

The <li> tag is used to create a list item.

I. Ordered List

The <ol> tag is used to create an ordered list. The type attribute specifies the type of ordering.

Type	Value
1	Numbers
A	Uppercase Letters
a	Lowercase Letters
I	Uppercase Roman Numerals
i	Lowercase Roman Numeral

II. Unordered List

The <ul> tag is used to create an unordered list.

III. Description List

The <dl> tag is used to create a description list. It is a list of terms and their descriptions.

Tag	Meaning
<dt>	Description Term
<dd>	Description Data

Creating Links

The `<a>` tag is used to create hyperlinks. A hyperlink connects different webpages together.

Attribute	Description
`href`	specifies the destination url of the hyperlink
`download`	creates a download link
`target="_blank"`	opens the link in a new tab

An **Absolute reference** contains the complete address of a resource.
eg. https://github.com/MohamedDevs
A **Relative reference** is the address of a resource relative to the current page.

Embedding Media

`<img>`

used to insert an image into the webpage.

Attribute	Description
`src`	specifies the path of the image
`alt`	displays an alternate text when the image fails to load
`width`	width of the image
`height`	height of the image
`loading="eager"`	loads immediately when the webpage loads
`loading="lazy"`	loads only when user scrolls near the image (best for images below the fold)

`<audio>`

used to insert audio into the webpage.

```
<audio controls>
      <source src="" type="audio/format">
</audio>
```

`<video>`

used to insert video into the webpage.

```
<video controls poster="">
      <source src="" type="video/format">
</video>
```

`poster` attribute of the `<video>` tag is used to insert a thumbnail for the video clip.

`<source>`

The source tag is used to specify the location of an audio/video clip.

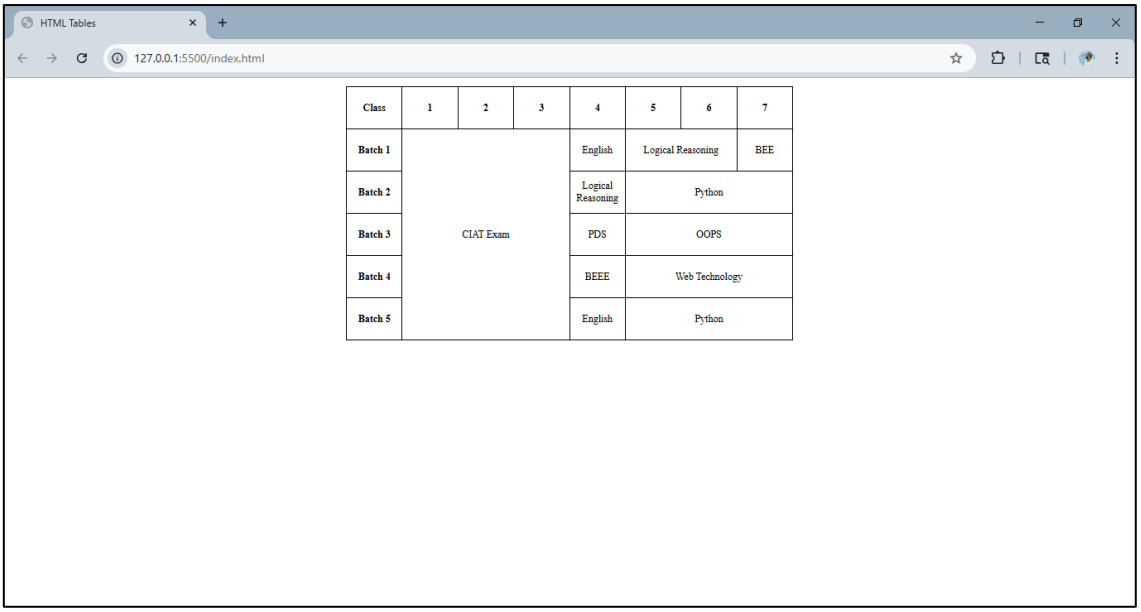
```html
<html lang="en">
    <head>
        <title>Embedding Media</title>
        <meta charset="UTF-8">
        <meta viewport="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <p>
            <img src="images/website-logo.png" alt="logo" width="150" height="150">
        </p>

        <p>
            <audio controls>
                <source src="yo_phone_linging.mp3" type="audio/mp3">
            </audio>
        </p>

        <p>
            <video controls width="200" height="200" poster="orangutan.jpg">
                <source src="monke.mp4" type="video/mp4">
            </video>
        </p>
    </body>
</html>
```

Creating Tables

Tag	Description
`<table>`	creates a table
`<tr>`	creates a table row
`<td>`	creates a table cell
`<th>`	creates a table header cell
`<td rowspan="x">`	makes the table cell span x rows
`<td colspan="y">`	makes the table cell span y columns
`<th scope="row">`	indicates the header is for a row
`<th scope="col">`	indicates the header is for a column

```html
<body>
    <table>
        <tr>
            <th scope="col">Class</th>
            <th scope="col">1</th>
            <th scope="col">2</th>
            <th scope="col">3</th>
            <th scope="col">4</th>
            <th scope="col">5</th>
            <th scope="col">6</th>
            <th scope="col">7</th>
        </tr>
        <tr>
            <th scope="row">Batch 1</th>
            <td rowspan="5" colspan="3">CIAT Exam</td>
            <td>English</td>
            <td colspan="2">Logical Reasoning</td>
            <td>BEE</td>
        </tr>
        <tr>
            <th scope="row">Batch 2</th>
            <td>Logical Reasoning</td>
            <td colspan="3">Python</td>
        </tr>
```

```html
        <tr>
            <th scope="row">Batch 3</th>
            <td>PDS</td>
            <td colspan="3">OOPS</td>
        </tr>
        <tr>
            <th scope="row">Batch 4</th>
            <td>BEEE</td>
            <td colspan="3">Web Technology</td>
        </tr>
        <tr>
            <th scope="row">Batch 5</th>
            <td>English</td>
            <td colspan="3">Python</td>
        </tr>
    </table>
</body>
</html>
```



HTML Forms

The `<form>` tag is used to create a form. It has two attributes:

- `action` – specifies the destination of form data
- `method` – specifies the HTTP method to be used while submitting form data
  - `GET` : form data is appended to the URL in name/value pairs
  - `POST` : form data is appended inside the body of the HTTP request

`<input>`

The `<input>` tag is used to get form input

- `type` – specifies the type of input
- `name` – key sent to the server
- `id` – links the input field to its `<label>`
- `placeholder` – used to create a placeholder text
- `required` – indicates a required field
- `value` – default value for a input field

Type	Description
`text`	single line of text input
`date`	opens a datepicker UI
`tel`	get a phone number input
`password`	creates a mask for input
`email`	get an email input
`number`	get numeric input. `max`,`min` can be used
`radio`	for selecting a single option
`checkbox`	for selecting multiple options

`<label>`

The `<label>` tag is used to create a label for an input field. Its `for` attribute must match with `id` attribute of the `input` tag.

<select>

The `<select>` tag is used to create a selectable list of options. Each of the options is created using the `<option>` tag.

<button>

This tag is used to create a clickable button. There are three types of button:

`submit` – create a button that submits form data

`reset` – create a button that clears form content

`button` – create a ordinary button

<fieldset>

The `<fieldset>` tag creates a border around the form. The `<legend>` tag is used to create a caption for the form.

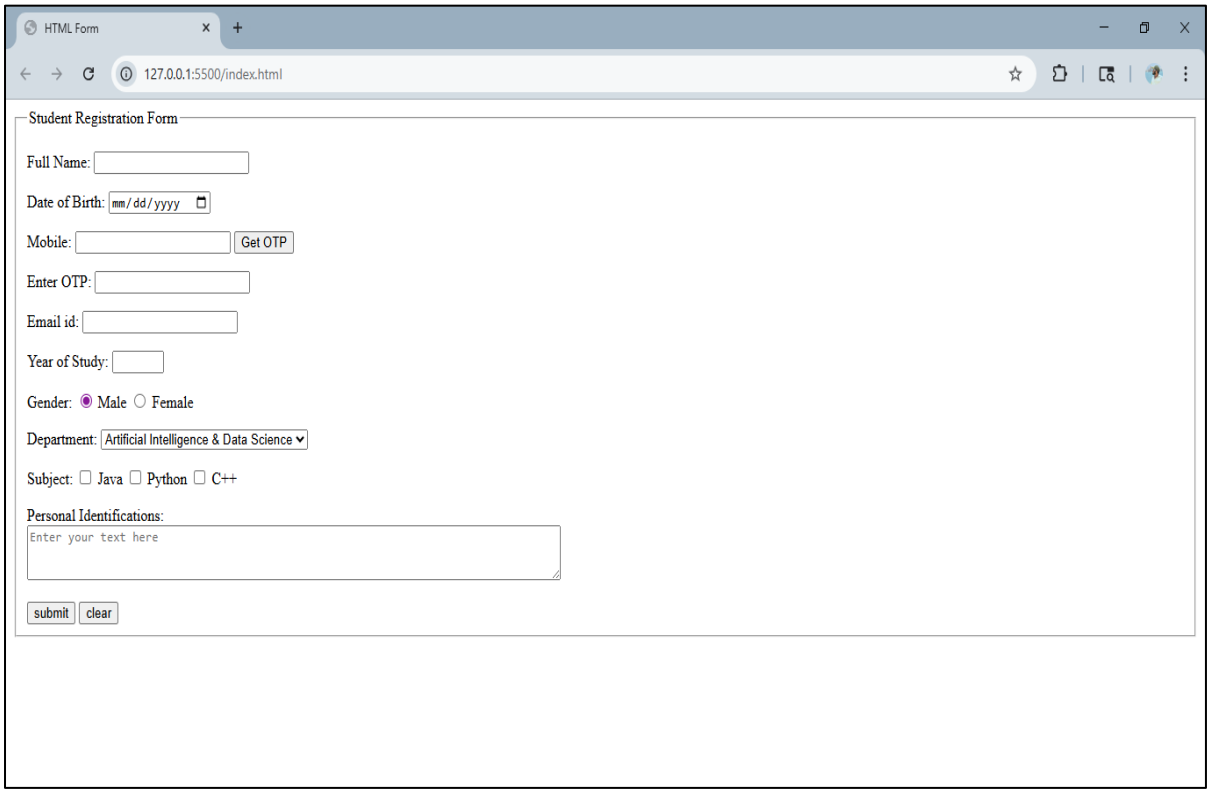
<textarea>

This tag is used to get multiple lines of input. Its `rows` and `cols` attribute are used to specify the number of rows and columns of the textarea respectively.

```html
<body>
    <fieldset><legend>Student Registration Form</legend>
    <form action="https://httpbin.org/post" method="POST">
        <p>
            <label for="fName">Full Name: </label>
            <input type="text" id="fName" name="fullName" required>
        </p>
        <p>
            <label for="dob">Date of Birth: </label>
            <input type="date" id="dob" name="DOB" required>
        </p>
        <p>
            <label for="ph">Mobile: </label>
            <input type="tel" id="ph" name="ph" required>
            <button type="button">Get OTP</button>
        </p>
        <p>
            Enter OTP:
            <input type="password" name="password" required>
        </p>
```

```html
<p>
    <label for="email">Email id: </label>
    <input type="email" id="email" name="email">
</p>
<p>
    <label for="year">Year of Study: </label>
    <input type="number" id="year" name="year" max="4" min="1">
</p>
<p>
    <label>Gender: </label>
    <input type="radio" id="male" name="gender" value="male" checked>
    <label for="male">Male</label>
    <input type="radio" id="female" name="gender" value="female">
    <label for="female">Female</label>
</p>
<p>
    <label for="dept">Department: </label>
    <select name="dept" id="dept">
        <option value="cse">Computer Science Engineering</option>
        <option value="aids" selected>Artificial Intelligence & Data Science</option>
        <option value="cyber">Cyber Security</option>
    </select>
</p>
```

```html
            <p>
                <label>Subject: </label>
                <input type="checkbox" name="subject[]" value="Java" id="java">
                <label for="java">Java</label>
                <input type="checkbox" name="subject[]" value="Python" id="python">
                <label for="python">Python</label>
                <input type="checkbox" name="subject[]" value="C++" id="cpp">
                <label for="cpp">C++</label>
            </p>
            <p>
                Personal Identifications:
                <br>
                <textarea name="pid" id="pid" rows="3" cols="80" placeholder="Enter your text here"></textarea>
            </p>
            <button type="submit">submit</button>
            <button type="reset">clear</button>
        </form>
        </fieldset>
    </body>
</html>
```



Block-level elements

Block-level elements take up the whole width and always begin in a new line.

eg. Elements like `<p>`, `<section>`, etc.

Inline elements

Inline elements only occupy necessary width and do not start on a new line.

eg. Elements like `<a>`, `<img>`, etc.

Semantic Tags

Semantic tags are elements that clearly describe their functionality.

<header>

Header part of the webpage

<main>

Main section of the webpage

<footer>

Footer part of the webpage

<article>

Creates an article inside the webpage

<section>

Creates a section inside the webpage

<nav>

Creates a navigation section that help users to navigate through the webpage. It contains links to different parts of the website or other pages.

<thead>

Head section of a table

<tbody>

Body section of a table

<tfoot>

Foot section of a table